Wentao Huang

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EDUCATION

National University of Singapore *Aug 2020 – Feb 2026*
- Ph.D. in Computer Science | GPA: 4.70/5.00
- Advisor: Prof. Kian-Lee Tan (ACM Fellow)

Renmin University of China *Sep 2017 – Jun 2020*
- M.Phil. in Computer Science | GPA: 3.74/4.00
- Advisor: Prof. Ju Fan

Shandong University *Sep 2013 – Jun 2017*
- B.Eng. in Electronic Engineering | GPA: 4.44/5.00 | Rank: 1/39
- Chinese Academy of Sciences (SITP) Elite Class

EXPERIENCE

School of Computing, National University of Singapore *Aug 2024 – Jun 2026*
- Student Research Assistant | Principal Investigator: Prof. Kian-Lee Tan (ACM Fellow)

Phancy Group Co, Ltd, Singapore *Aug 2023 – Apr 2025*
- Ph.D. Student Researcher | Mentors: Dr. Mian Lu, Prof. Bingsheng He (IEEE Fellow)

Illinois Advanced Research Center at Singapore *Jul 2018 – Feb 2019*
- Research Engineer | Mentors: Prof. Yuchen Li, Prof. Yuan Fang

PUBLICATIONS

Conference Papers

[C6] **Wentao Huang**, Qiming Huang, Yunhong Ji, Mian Lu, Yuqiang Chen, Bingsheng He, Kian-Lee Tan. "Taming Multi-Dimensional Skew in Sparse Matrix Multiplication with Contention-Aware Scheduling." 53rd International Conference on Very Large Data Bases (VLDB), 2027.

[C5] **Wentao Huang**, Fanfan Lin. "Quantum-Enhanced Knowledge Graph Systems: Survey, Taxonomy, and the Road to Scalability." 7th International Conference on Quantum Computing and Engineering (QCE), 2026.

[C4] **Wentao Huang**, Mian Lu, Kian-Lee Tan. "Hash Joins Meet CXL: A Fresh Look." 16th Conference on Innovative Data Systems Research (CIDR), 2026.

[C3] Yunhong Ji, **Wentao Huang**, Xuan Zhou, Bingsheng He, Kian-Lee Tan. "TaC: An Anti-Caching Key-Value Store on Heterogeneous Memory Architectures." 27th International Conference on Extending Database Technology (EDBT), 2024.

[C2] **Wentao Huang**, Yunhong Ji, Xuan Zhou, Bingsheng He, Kian-Lee Tan. "A Design Space Exploration and Evaluation for Main-Memory Hash Joins in Storage Class Memory." 49th International Conference on Very Large Data Bases (VLDB), 2023.

[C1] **Wentao Huang**, Yuchen Li, Yuan Fang, Ju Fan, Hongxia Yang. "BiANE: Bipartite Attributed Network Embedding." 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2020.

Journal Papers

[J2] Tongfeng Weng, Mo Sha, Xu Zhou, Jingjing Lu, **Wentao Huang**, Kenli Li, Kian-Lee Tan. "SMDG: Enhancing In-Memory Dynamic Graph Processing With Storage-Class Memory." IEEE Transactions on Knowledge and Data Engineering (TKDE), 2026.

[J1] Yunhong Ji, **Wentao Huang**, Xuan Zhou. "HeterMM: applying in-DRAM index to heterogeneous memory-based key-value stores." Frontiers of Computer Science (FCS), 2024.

Workshop Papers

[W1] **Wentao Huang**, Mo Sha, Mian Lu, Yuqiang Chen, Bingsheng He, Kian-Lee Tan. "Bandwidth Expansion via CXL: A Pathway to Accelerating In-Memory Analytical Processing." 15th International Workshop on Accelerating Analytics and Data Management Systems Using Modern Processor and Storage (ADMS), 2024.

INVITED TALKS

Scaling Data Infrastructure: Building Bandwidth-Intensive Systems with CXL. *Aug 2026*
– Institute of Computing Technology, Chinese Academy of Sciences

Accelerating Data-Intensive Systems with Emerging Large-Scale Memory Technologies. *Dec 2025*
– Shanghai Innovation Institute

Efficient Analytical Query Processing with Emerging Large-Scale Memory Technologies. *Dec 2025*
– Beihang University

PROFESSIONAL SERVICE

Review Board / Program Committee Member | VLDB'27, ADMS'25, DL4C@NeurIPS'25 *2025, 2027*

Invited Reviewer | FCS'26, DPD'26, ISCAS'24 *2024, 2026*

Student Area Search Committee | NUS School of Computing *2025, 2026*

Research Session Deputy Chair | EDBT'24 *2024*

Artifact Evaluation Committee | SOSP'23, OSDI'23, EuroSys'23, USENIX ATC'23 *2023*

Shadow Program Committee Member | VLDB'26 *2026*

PROJECTS

Apache SINGA *Jul 2018 – Aug 2018*
– Mentors: Dr. Wei Wang, Prof. Ju Fan | Source code: github.com/apache/singa

TEACHING

CS3223: Database Systems Implementation *Jan 2021 – May 2022*
Teaching Assistant, National University of Singapore

Linear Algebra *Feb 2019 – Jun 2019*
Teaching Assistant, Renmin University of China

Introduction to Computer Science *Feb 2018 – Jun 2018*
Teaching Assistant, Renmin University of China

AWARDS & PRIZES

Research Achievement Award, National University of Singapore *Aug 2023 – Jul 2024*

The NUS Research Scholarship, National University of Singapore *Aug 2020 – Jul 2024*

The Postgraduate Scholarship, Renmin University of China *Sep 2017 – Jun 2020*

The Undergraduate Scholarship, Shandong University *Sep 2013 – Jun 2017*

"Challenge Cup" Science & Technology Competition, Gold Award, Shandong University *May 2016*

Innovation and Entrepreneurship Competition, First Prize, Shandong Province *Oct 2015*

The Chinese Mathematics Competition, Third Prize, China *Mar 2015*

National Undergraduate Electronic Design Contest, Second Prize, Shandong Province *Sep 2014*

University Football Cup, Champion, Shandong University *Apr 2014*

MISCELLANEOUS

Open Water Diver, Scuba Schools International *Sep 2018 – Present*

Moderator of Technical Q&A, TensorFlow China Community *Sep 2018 – Aug 2019*

Minister of English Broadcasting Station, Shandong University *Aug 2014 – Jul 2015*